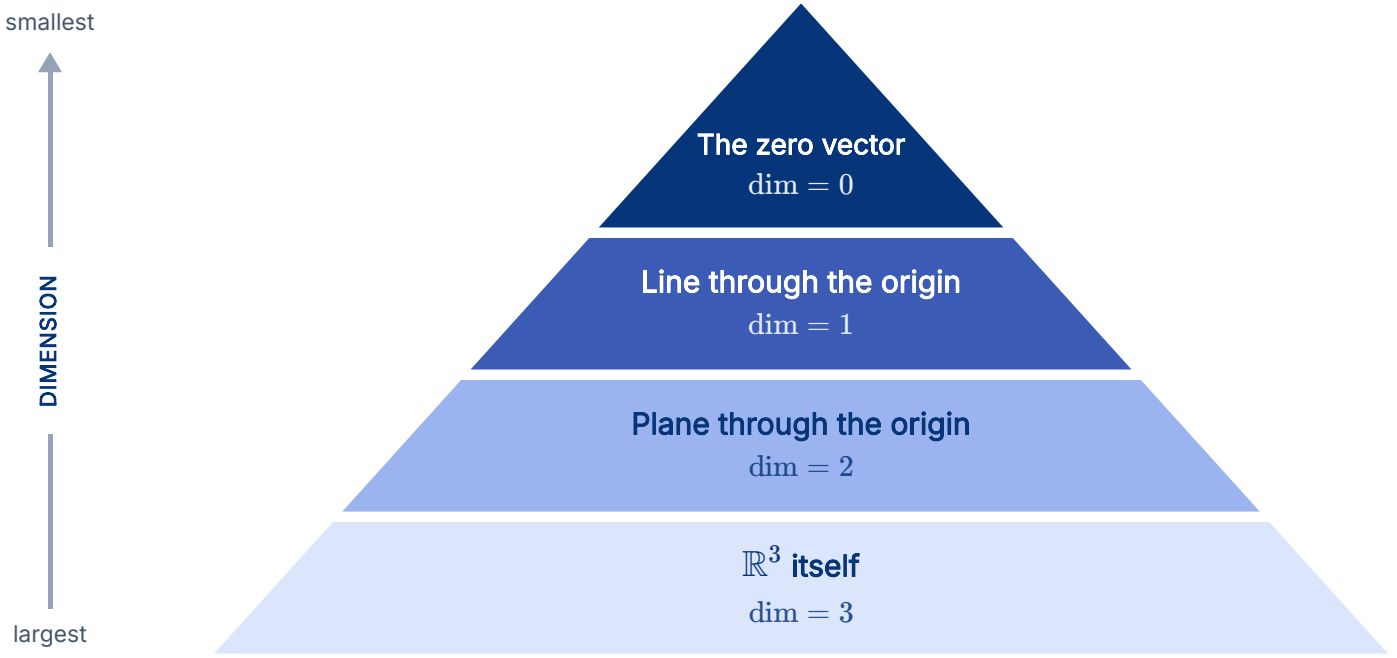
BEFORE YOU READ

Subspaces of $\mathbb{R}^3$ by dimension

If W is a subspace of V, then $\dim(W) \leq \dim(V)$, with equality only when $W = V$. In $\mathbb{R}^3$ that leaves exactly four possible dimensions.



TAKEAWAY

Always a whole number

There is no subspace of dimension $\frac{3}{2}$ or π: dimension is always a non-negative integer. See subspaces for these four kinds.